



What Is a Variable?

Name: _____

Date: _____

Class Period: _____

1 WARM-UP

Goal: Translate keywords into mathematical symbols and evaluate simple expressions. Grades are earned on a locked 4-level scale: Emerging → Developing → Proficient → Superlative. Proficient is the passing level; Superlative earns an A.

💡 Hint: Variables are letters (like x , n , p) that represent unknown quantities or values that can change.

📅 Unit 1 Rubric: How Grades Are Earned

Benchmark	Emerging	Developing	Proficient	Superlative
9.3.5.3	Recognizes but cannot yet apply number systems.	Applies number systems with support or on routine problems only.	Correctly applies number systems to solve problems and explain reasoning.	Independently applies number systems to novel problems and justifies the most efficient approach.
9.3.5.5	Recognizes but cannot yet apply cost, discounts, taxes & order of operations.	Applies cost, discounts, taxes & order of operations with support or on routine problems only.	Correctly applies cost, discounts, taxes & order of operations to solve problems and explain reasoning.	Independently applies cost, discounts, taxes & order of operations to novel problems and justifies the most efficient approach.
9.3.5.7	Recognizes but cannot yet apply equivalent forms & rearranging formulas.	Applies equivalent forms & rearranging formulas with support or on	Correctly applies equivalent forms & rearranging formulas to solve problems and explain reasoning.	Independently applies equivalent forms & rearranging formulas to novel problems and

Benchmark	Emerging	Developing	Proficient	Superlative
		routine problems only.		justifies the most efficient approach.
9.3.5.8	Recognizes but cannot yet apply writing expressions in multiple ways.	Applies writing expressions in multiple ways with support or on routine problems only.	Correctly applies writing expressions in multiple ways to solve problems and explain reasoning.	Independently applies writing expressions in multiple ways to novel problems and justifies the most efficient approach.

1. Write an algebraic expression for “2 more than x .”

• Expression: _____

2. Write an algebraic expression for “5 times b .”

• Expression: _____

3. Write an algebraic expression for “10 less than y .”

• Expression: _____

4. Write an algebraic expression for “ a divided by 3.”

• Expression: _____

5. Evaluate $x + 4$ when $x = 6$.

• Show your work: _____ * Value: _____

6. Evaluate $3n$ when $n = 7$.

• Show your work: _____ * Value: _____

2 VOCABULARY — MATCH THE TERM TO ITS DEFINITION

Use the word bank. Match each term to its symbol or definition and write the matching letter in the box. Word Bank

✓	Keyword	Symbol / Operation
_____	sum	+
_____	difference	—
_____	product	$\times$ or $()$

✓	Keyword	Symbol / Operation
—	quotient	$\div$ or fraction bar
—	twice	$\times 2$

3 GUIDED PRACTICE

Goal: Write expressions for more complex phrases and evaluate them. 💡 Hint: When writing an expression, pay attention to the order of words. “Three less than a number” means you start with the number and subtract three ($n - 3$), not $3 - n$.

1. Write an algebraic expression for “twice a number increased by 5.”

• Expression: _____

2. Write an algebraic expression for “8 fewer than 3 times k .”

• Expression: _____

3. Write an algebraic expression for “the product of m and 9.”

• Expression: _____

4. Evaluate $4x - 2$ when $x = 9$.

• Show your work: _____ * Value: _____

5. Evaluate $y \div 2 + 3$ when $y = 14$.

• Show your work: _____ * Value: _____

6. The sum of a number and 12 equals 25. Write and solve an equation.

• Equation: _____ * Value: _____

4 PARTNER PRACTICE

Sentence frames: First, I _____. Then, I _____. Finally, I _____ because _____.

Goal: Solve equations and evaluate more complex expressions. 💡 Hint: To solve an equation, undo each operation in reverse order (GEMDAS backward).

1. Evaluate $6n - 4$ when $n = 8$.

• Show your work: _____ * Value: _____

2. Write an algebraic expression for “the quotient of a number and 4, plus 9.”

• Expression: _____

3. Solve: $4n + 6 = 22$

• Show your work: _____ * Value: _____

4. Seven more than 3 times a number is 28. Write and solve an equation.

• Equation: _____ * Value: _____

5. Evaluate $5x - x$ when $x = 6$.

• Show your work: _____ * Value: _____

6. Twice a number decreased by 5 equals 13. Write and solve an equation.

• Equation: _____ * Value: _____

SELF-ASSESSMENT

Rate yourself on the following criteria using the scale below.

Scale:

4 = Independent | 3 = Accurate | 2 = Needs Support | 1 = Still Learning

Criteria	Rating (4-3-2-1)	Reflection / Goal
I can define a variable and identify it in expressions.		
I can translate keywords (sum, difference, product, quotient) into symbols.		
I can select variables and write expressions for real-world scenarios.		
I can create models and reason about variable constraints/interpretations.		

Performance Descriptors:

- 4: Can define, identify variables, and write complex expressions; creates accurate real-world models.
- 3: Can define, identify variables, and write basic expressions accurately.
- 2: Can identify variables but needs support with translation or writing expressions.
- 1: Is still learning what a variable is and how to represent unknowns.